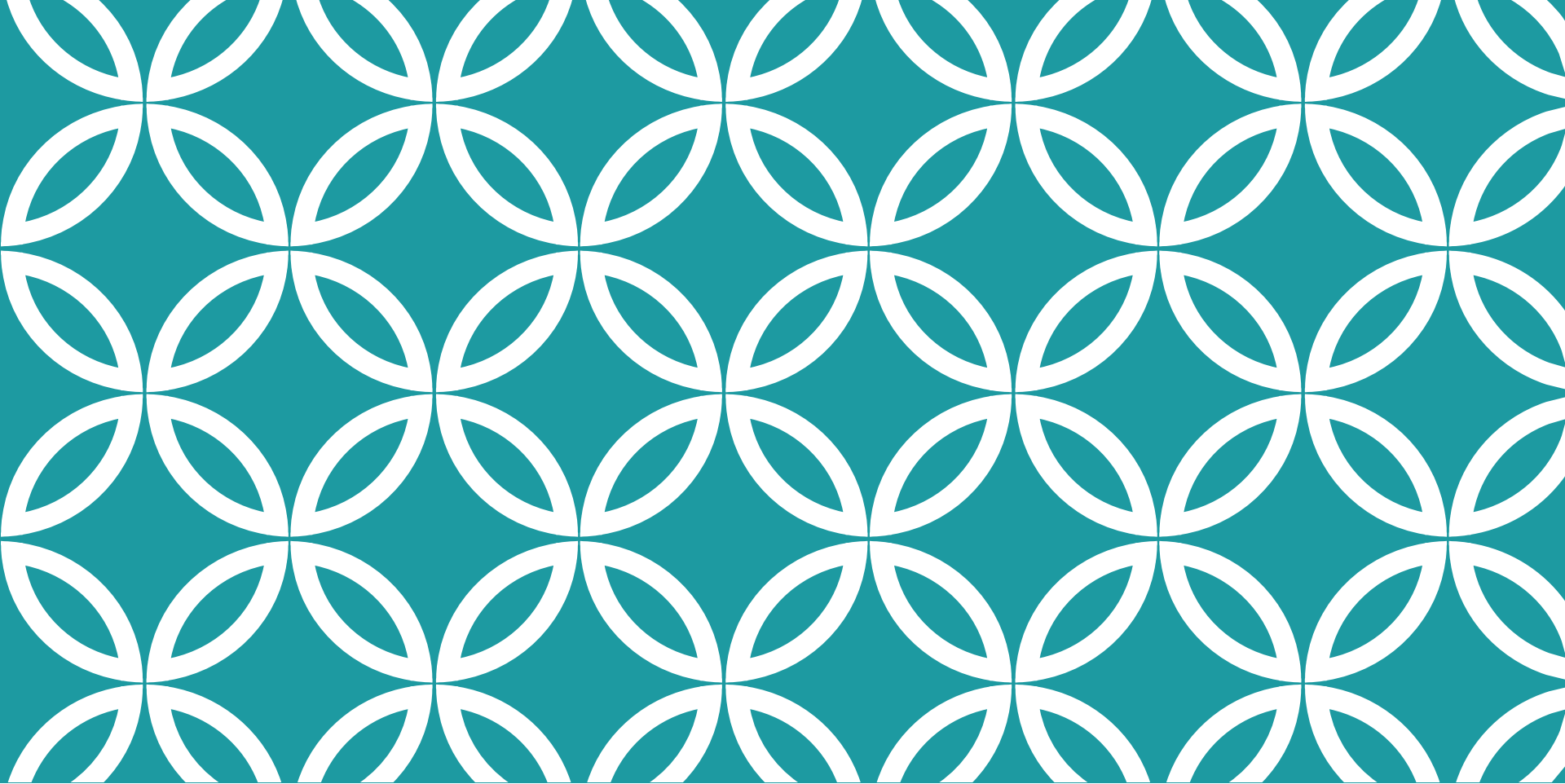


LECTURE2: SPEECH PRODUCTION & PERCEPTION, CHARACTERISTICS OF SPEECH SIGNAL

Reference Text Books:

1. Fundamentals of Speech recognition – L. Rabiner and B. Juang, Prentice Hall signal processing series.
2. Discrete-time speech signal processing: principles and practice, Thomas F. Quatieri, Coth.



SPEECH PRODUCTION & PERCEPTION



SPEECH PRODUCTION AND PERCEPTION

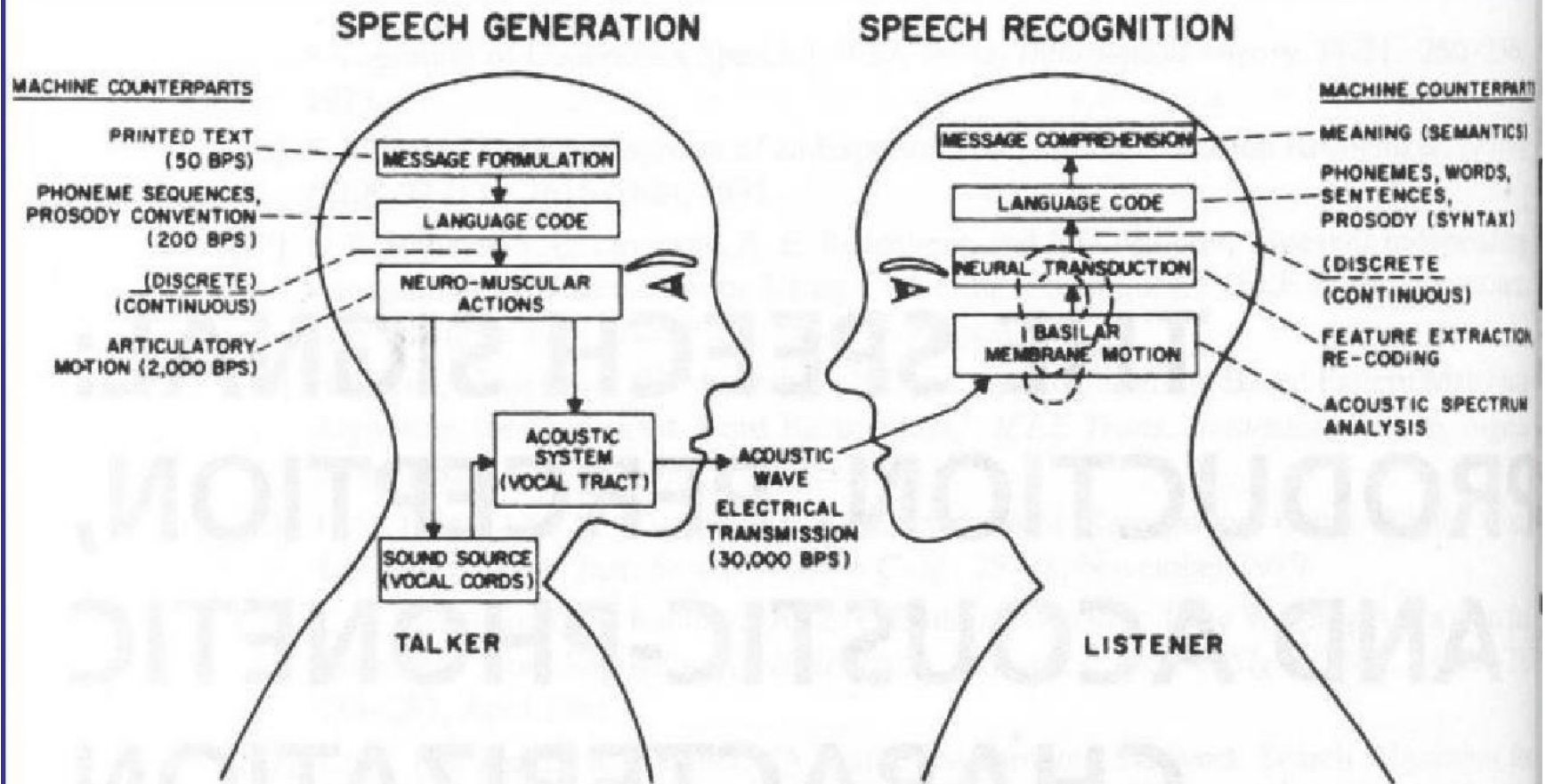


Figure 2.1 Schematic diagram of speech-production/speech-perception process (after Flanagan [unpublished]).

SPEECH PRODUCTION

The production process (generation) begins when the talker formulates a message in his mind which he wants to transmit to the listener via speech.

In case of machine,

- First, the message is formed in terms of printed text.
- Next, the message is converted into a language code. This roughly corresponds to converting the text of the message into a set of phoneme sequences, along with the prosody markers denoting duration, loudness, pitch accent of the sound.

After the language code is chosen the talker must execute a series of neuromuscular commands to cause the vocal cord to vibrate such that the proper sequence of speech sounds is created.

The neuromuscular commands must simultaneously control the movement of lips, jaw, tongue, and velum.

SPEECH PRODUCTION MECHANISM

Vocal tract consist of

- Pharynx: the connection from the esophagus to the mouth
- Mouth or oral cavity

The nasal tract begins at the velum and ends at the nostrils.

Velum is a trap-door like mechanism at the back of the mouth cavity. When the velum is lowered, the nasal tract is acoustically coupled to the vocal tract to produce nasal sounds of speech.

The lungs and the associated muscles excites the vocal mechanism.

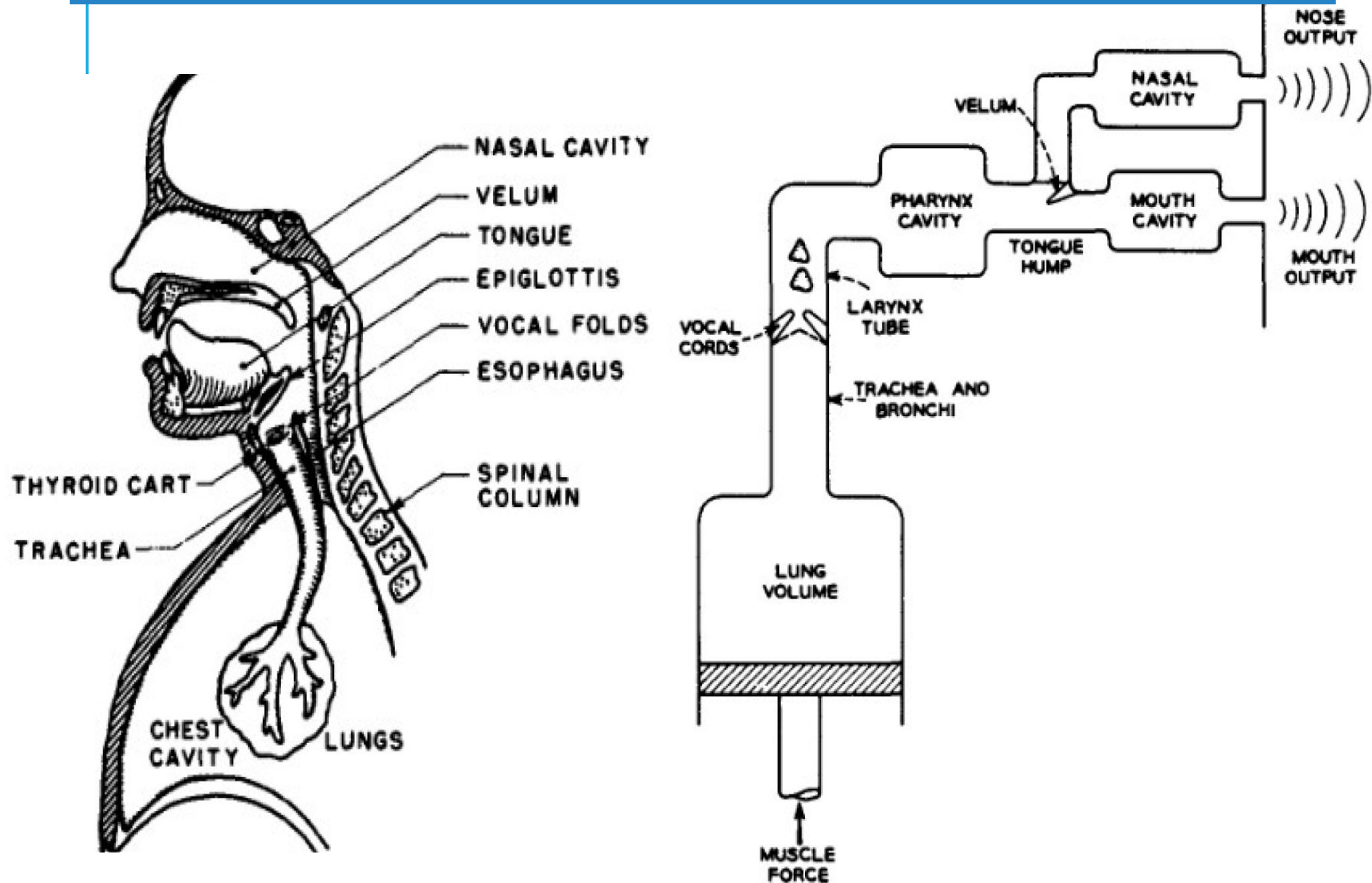
The muscle force pushes air out of the lungs and through the bronchi and trachea.

When the vocal cord is tensed, the air flow causes them to vibrate, produces so called voice-speech sounds.

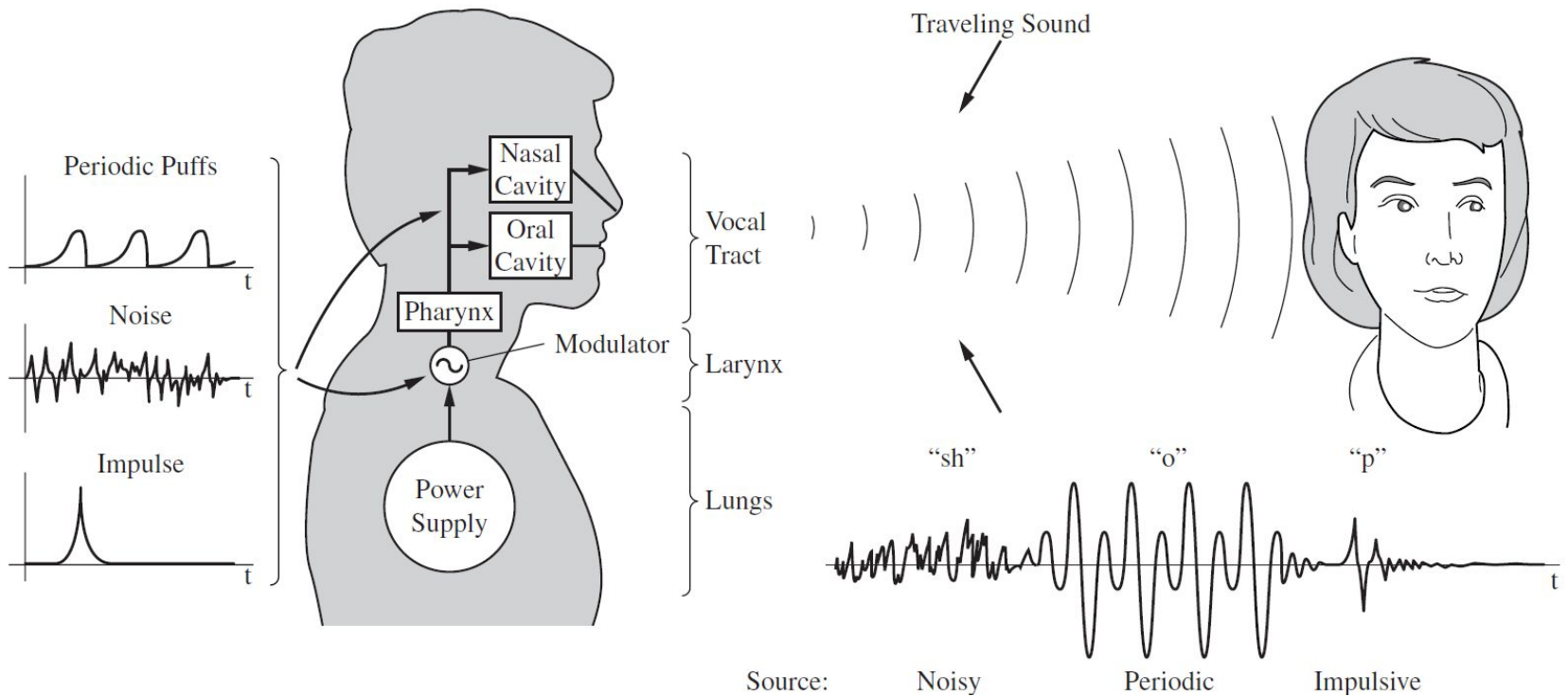
When the vocal cord is relaxed, a sound is produced.

Speech is produced as a sequence of sounds.

HUMAN VOCAL MECHANISM



SPEECH PRODUCTION



SPEECH PERCEPTION

The speech signal is generated and propagated to the listener, the speech perception(recognition) process begins.

First the listener processes the acoustic signal along the basilar membrane in the inner ear, which provides running spectral analysis of the incoming signal.

A neural transduction process converts the spectral signal at the output of the basilar membrane into activity signals on the auditory nerve., corresponding roughly to a feature extraction process.

The neural activity along the auditory nerve is converted to a language code at the higher centers of processing brain and finally, message comprehension is achieved.

The audible frequency range for human is appx 20Hz to 20KHz.

SPEECH PERCEPTION

The three distinct parts of the ear are outer ear, middle ear and inner ear.

- Outer ear:

- Outer ear includes pinna and auditory canal. It helps to direct an incident sound wave into middle ear. Filters and modifies the captured sound.
- The perceived sound is sensitive to the pinna's shape.
- By changing the pinna's shape the sound quality alters as well as background noise.

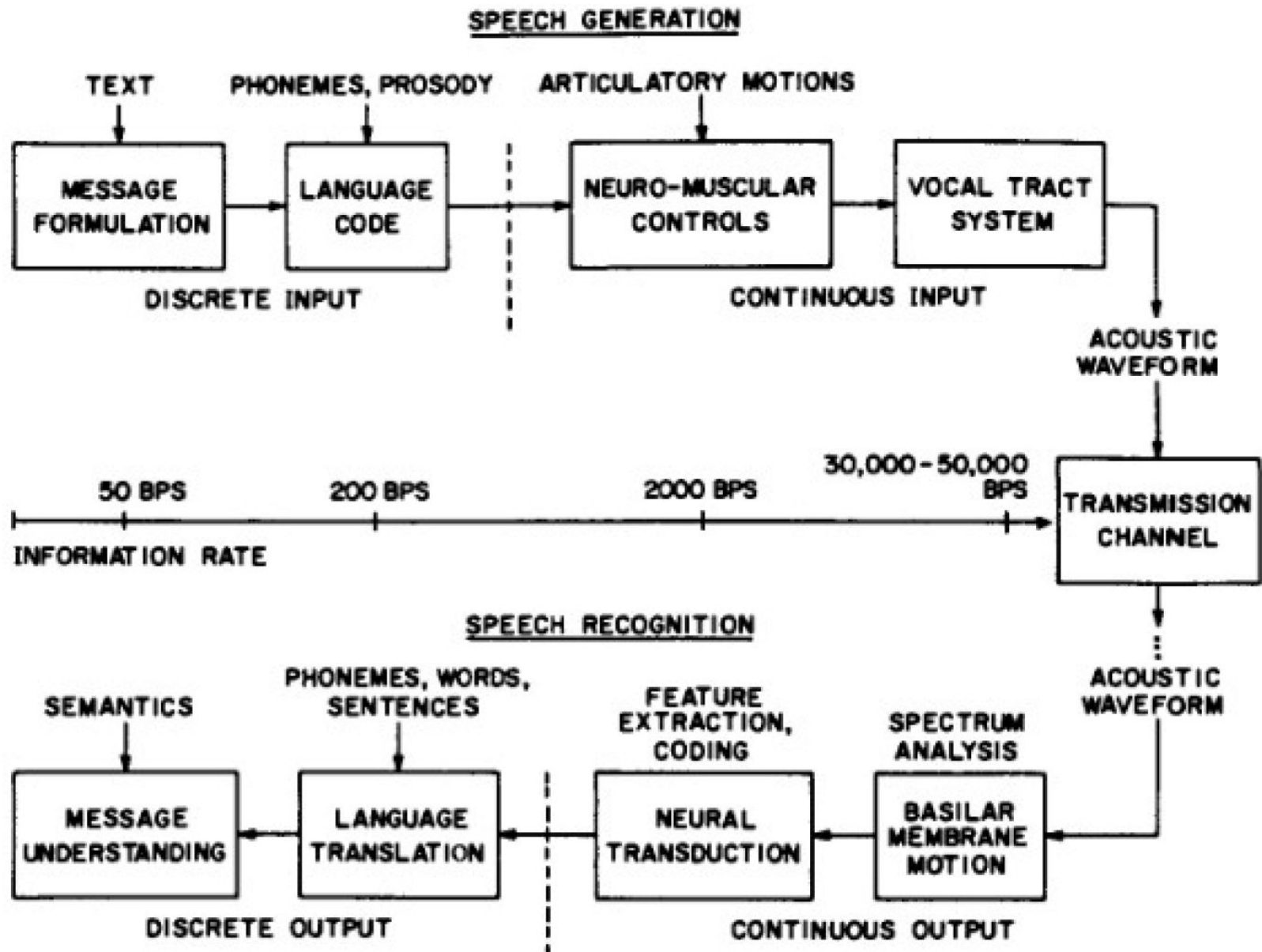
- Middle ear - Ear drum:

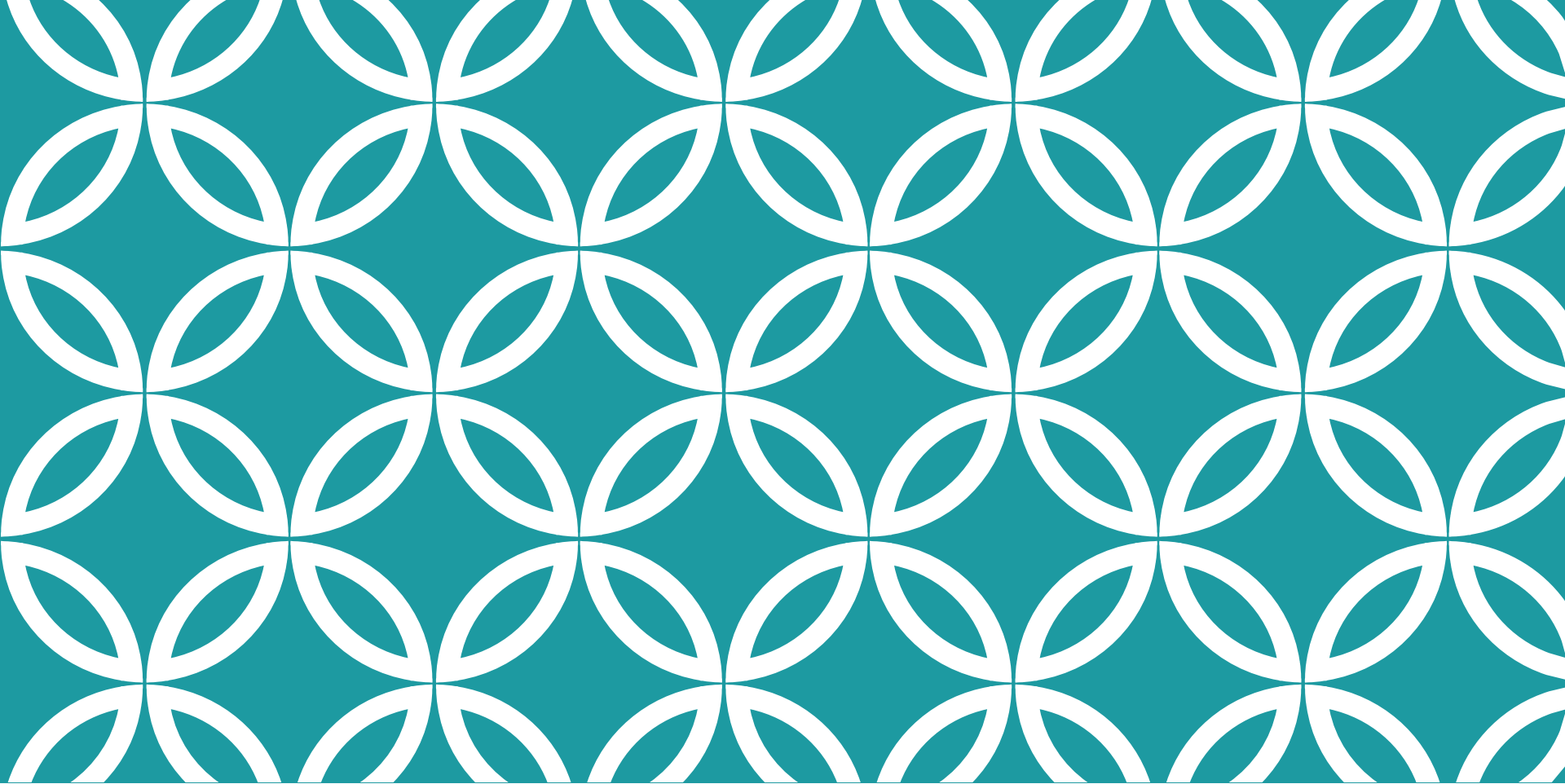
- After passing through ear canal sound wave strikes the eardrum which is part of middle ear.
- This oscillates with the frequency as that of the sound wave
- Movements of this membrane are then transmitted through the system of small bones called as ossicular system
- From ossicular system to cochlea

- Inner ear:

- Cochlea, a spiral cavity filled with fluid, converts sound waves into neural signals via hair cells.
- Provides accurate encoding of speech sounds
- Active processes in the cochlea, supported by neural feedback loops, enhance selective attention to speech and contribute to the “cocktail party effect” (focusing on one speaker among many).

ALTERNATE VIEW OF SPEECH PRODUCTION AND PERCEPTION





SPEECH SOUNDS AND FEATURES/CHARACTERISTICS

PHONEMES - IPA AND ARPABET

The number of linguistically distinct speech sounds in a language, also known as phonemes, is often a matter of judgement and is not invariant to different linguists.

IPA (International Phonetic Alphabet) is a standardized system of symbols used by linguists to represent the sounds of spoken language. It was created by the *International Phonetic Association* to provide a universal way of transcribing speech sounds across all languages. IPA is considered as the global “alphabet of sounds” that makes speech transcription precise and universal.

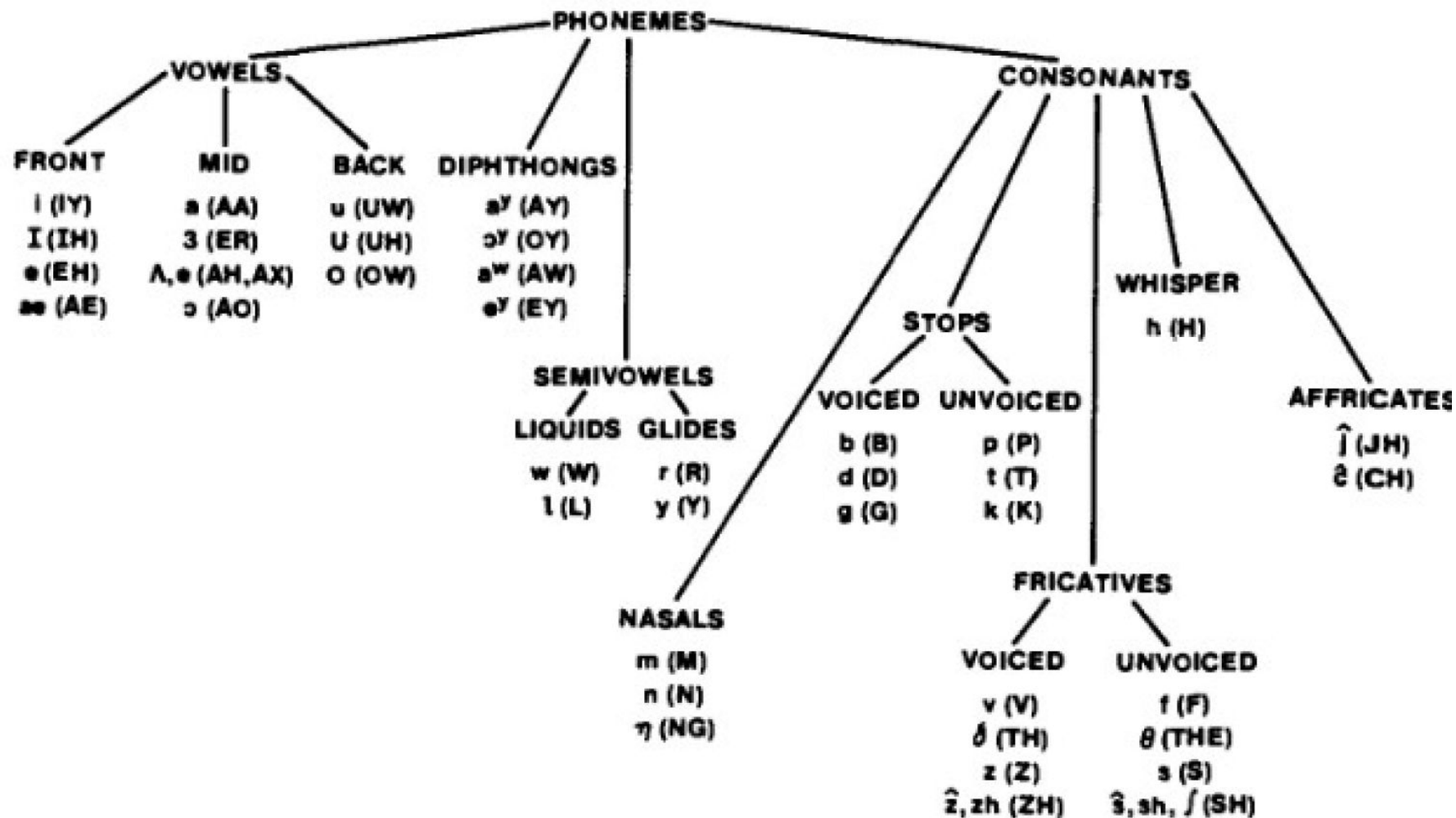
ARPAbet is a phonetic transcription system developed by ARPA (Advanced Research Projects Agency) for representing speech sounds in computer-readable form. It was widely used in early speech recognition and synthesis systems, especially in the U.S.

It is ASCII-based. Uses plain text characters instead of IPA symbols, making it easy for computers to process.

PHONEMES

Phoneme	ARPABET	Example	Phoneme	ARPABET	Example
/ɪ/	IY	beat	/ŋ/	NX	sing
/ʊ/	IH	bit	/p/	P	pet
/e/ (eʲ)	EY	bat	/t/	T	ten
/ɛ/	EH	bet	/k/	K	kit
/æ/	AE	bat	/b/	B	bet
/ɑ/	AA	Bob	/d/	D	debt
/ʌ/	AH	but	/g/	H	get
/ɔ/	AO	bought	/h/	HH	hat
/o/ (oʷ)	OW	boat	/f/	F	fat
/ʊ/	UH	book	/θ/	TH	thing
/u/	UW	boot	/s/	S	sat
/ə/	AX	about	/ʃ/ (sh)	SH	shut
/ɪ/	IX	roses	/v/	V	vat
/ɜ/	ER	bird	/ð/	DH	that
/ə/	AXR	butter	/z/	Z	zoo
/ɑʷ/	AW	down	/ʒ/ (zh)	ZH	azure
/aʲ/	AY	buy	/tʃ/ (tsh)	CH	church
/ɔʲ/	OY	boy	/dʒ/ (dzh, j)	JH	judge
/y/	Y	you	/w/	WH	which
/w/	W	wit	/l/	EL	battle
/r/	R	rent	/m/	EM	bottom
/l/	L	let	/n/	EN	button
/m/	M	met	/ŋ/	DX	batter
/n/	N	net	/ʔ/	Q	(glottal stop)

PHONEMES



PHONEMES

		IPA Symbol	ARPAbet (SV)	ARPAbet (UV)	Examples
Vowels	Front	i	i	IY	beat
		I	I	IH	bit
		e	e	EY	bait
		ɛ	E	EH	bet
		æ	@	AE	bat
	Back	ɑ	a	AA	Bob
		ɔ	c	AO	bought
		o	o	OW	boat
		U	U	UH	book
		u	u	UW	boot
	Mid	ɜ	R	ER	bird
		ə	x	AX	ago
		ʌ	ʌ	AH	but
Diphthongs		aI	Y	AY	buy
		aU	W	AW	down
		ɔI	O	OY	boy
		ɪ	X	IX	roses
Semivowels	Liquids	l	l	L	love
		ɫ	L	EL	cattle
		r	r	R	race
	Glides	w	w	W	want
		ʍ	H	WH	when
		j	y	Y	yard

PHONEMES

		IPA Symbol	ARPAbet (SV)	ARPAbet (UV)	Examples
Stop Consonants	Voiced	b	b	B	bat
		d	d	D	deep
		g	g	G	go
	Unvoiced	p	p	P	pea
		t	t	T	tea
		k	k	K	kick
Fricatives	Voiced	v	v	V	vice
		ð	D	DH	then
		z	z	Z	zebra
		ʒ	Z	ZH	measure
	Unvoiced	f	f	F	five
		θ	T	TH	thing
		s	s	S	so
		ʃ	S	SH	show
Nasal	Non vocalic	m	m	M	mom
		n	n	N	noon
		ŋ	G	NX	sing
	Vocalic	ɱ	M	EM	some
		ɯ	H	WH	when
		j	y	Y	yard

VOWELS

Th_y n_t_d s_gn_f_c_nt _mpr_v_m_nt _n th__r p_rf_rm_nc_d_r_ng
pl_c_m_nt

They noted significant improvement in their performance during placement

__e__o__e__i__i__i__a__i__o__e__e__i__ei__e__o__a__e__u__i__
__a__e__e__

Vowels are speech sounds produced with an open vocal tract, allowing air to flow freely, typically voiced and classified by tongue position (height and backness), lip rounding, and tenseness.

Vowels are generally long in duration compared to consonants.

They are usually easily and reliably recognized playing significant role for speech recognition, both by human beings and machines.

VOWELS

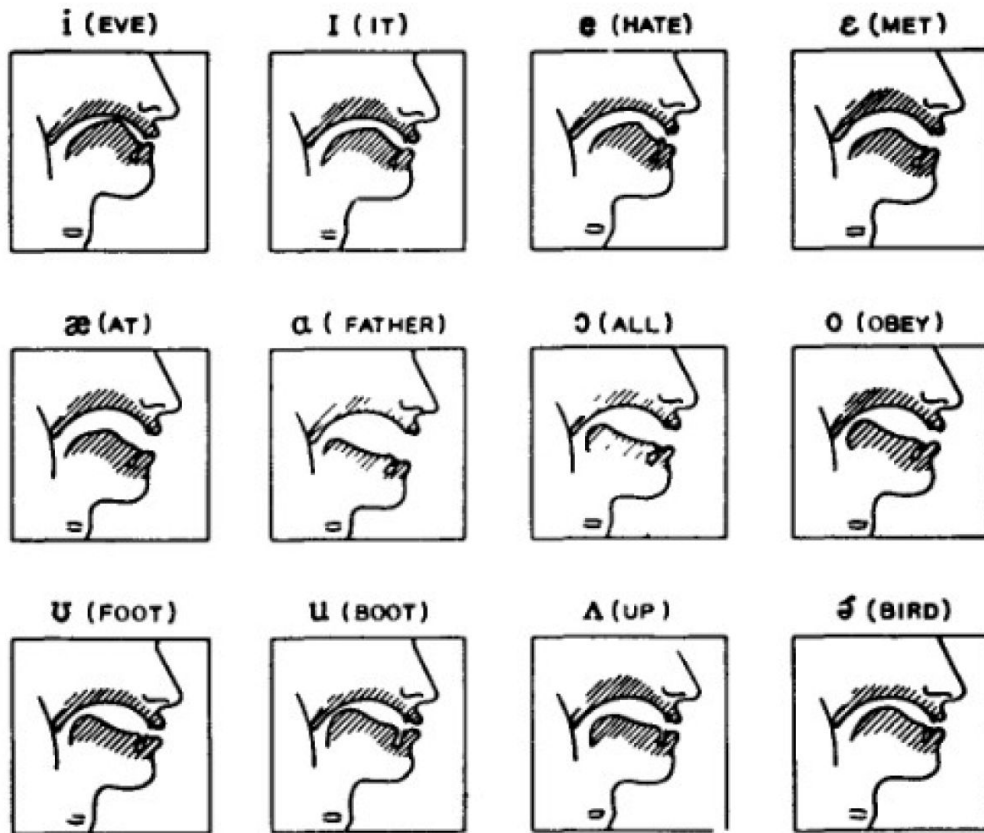
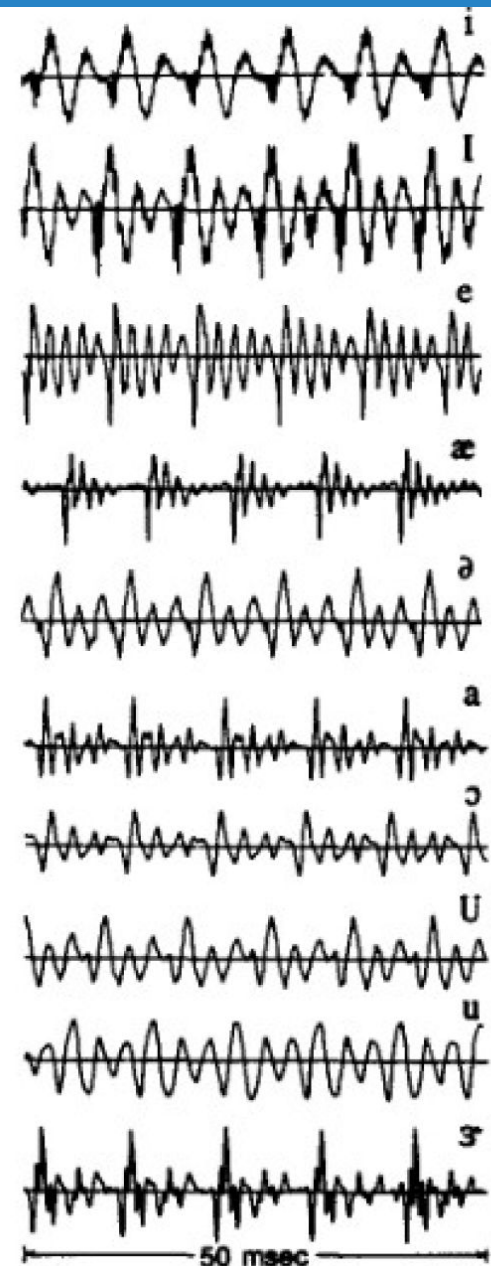


Figure 2.13 Articulatory configurations for typical vowel sounds (after Flanagan [3])



Acoustic Waveform plots of typical vowel sounds

VOWELS

Formant positions are the resonant frequency peaks of the vocal tract that shape vowel quality and distinguish speech sounds.

The first two formants (F1 and F2) are most critical: F1 corresponds inversely to tongue height, while F2 reflects tongue frontness/backness. Higher formants (F3, F4) add detail to timbre and speaker identity

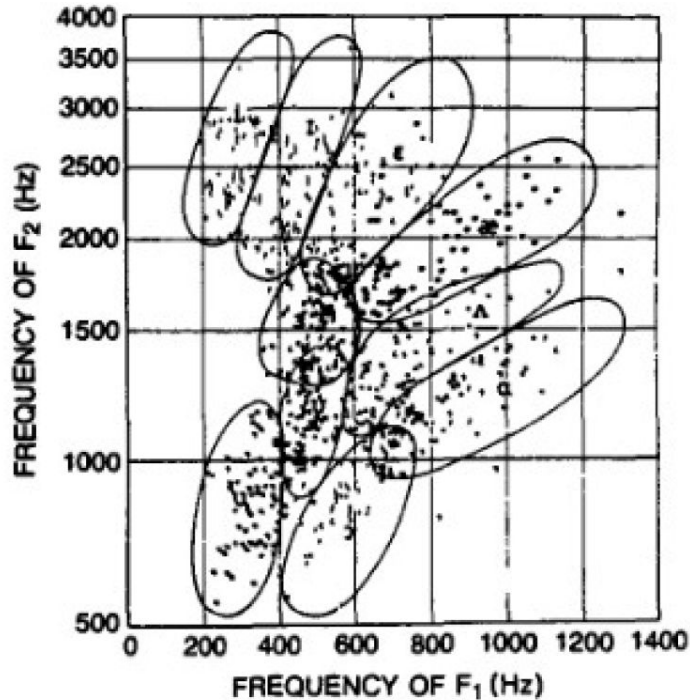


Figure 2.16 Measured frequencies of first and second formants for a wide range of talkers for several vowels (after Peterson & Barney [7])

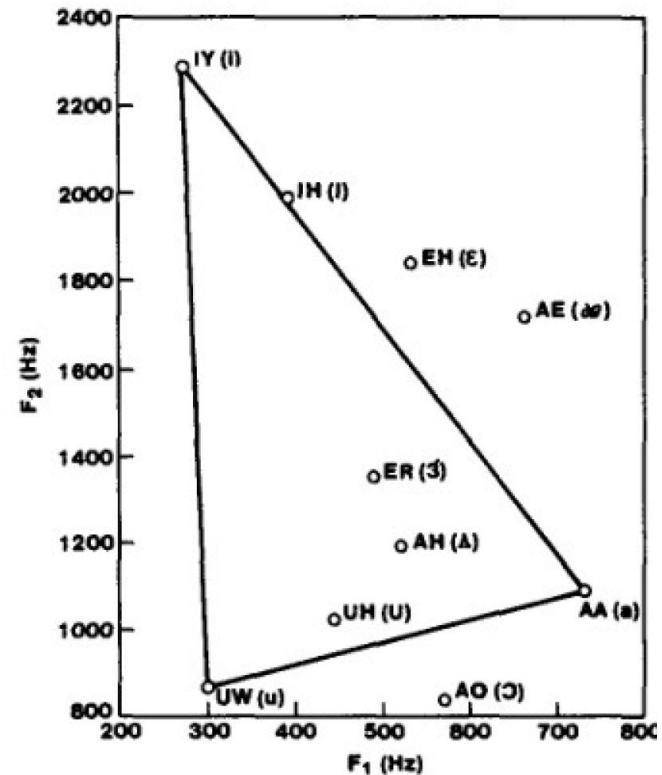


Figure 2.17 The vowel triangle with centroid positions of the common vowels.

DIPHTHONGS AND SEMI-VOWELS

A **diphthong** is a *gliding vowel sound* where the tongue moves from one position to another within the same syllable.

They start with one vowel and end with another, creating a smooth transition.

Examples in English:

/aɪ/ (*bite*) → from /a/ to /ɪ/

/eɪ/ (*make*) → from /e/ to /ɪ/

/ɔɪ/ (*boy*) → from /ɔ/ to /ɪ/

/aʊ/ (*house*) → from /a/ to /ʊ/

/oʊ/ (*boat*) → from /o/ to /ʊ/

Semi-vowels (also called **glides**) are sounds that behave like vowels acoustically but function like consonants in syllable structure.

Examples in English:

/j/ (*yes*) → similar to vowel /i/

/w/ (*we*) → similar to vowel /u/

CONSONANTS

Nasal Consonants - Air passes through the nose while the oral cavity is blocked.

Examples:

/m/ (man) – bilabial nasal

/n/ (net) – alveolar nasal

/ŋ/ (sing) – velar nasal

Fricatives - Produced by forcing air through a narrow constriction, creating turbulence.
Can be voiced or voiceless.

Examples:

Voiceless: /f/ (fun), /s/ (sun), /ʃ/ (shoe)

Voiced: /v/ (van), /z/ (zoo), /ʒ/ (measure)

Stops (Plosives) - Complete closure of the vocal tract followed by a sudden release of air.
Can be voiced or voiceless.

Examples:

Voiceless: /p/ (pat), /t/ (top), /k/ (cat)

Voiced: /b/ (bat), /d/ (dog), /g/ (go)